MySQL Interactive Firmware Development

Documentation:

1- Setup ESP32 on ArduinoIDE:

Step 1: Install the ESP32 Board Manager URL

- Open Arduino IDE
- Go to File → Preferences (on Windows/Linux)
- In the Additional Board Manager URLs field, paste the following URL:

http://arduino.esp8266.com/stable/package_esp8266com_index.json,https://dl.espressif.com/dl/package_esp32_index.json,https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

Step 2: Install the ESP32 Boards

- Go to Tools → Board → Boards Manager.
- In the search bar, type esp32.
- Find and select the esp32 by Espressif Systems package.
- Click Install and wait for the installation to complete

Step 3: Select the ESP32 Board

- Go to Tools → Board.
- Scroll down and select the appropriate ESP32 board (e.g., ESP32 Dev Module).

Step 4: Select the COM port.

- Go to Tools → Port.
- Select the port corresponding to your ESP32 board (e.g., COM3

NOTE: Arduino Libraries:

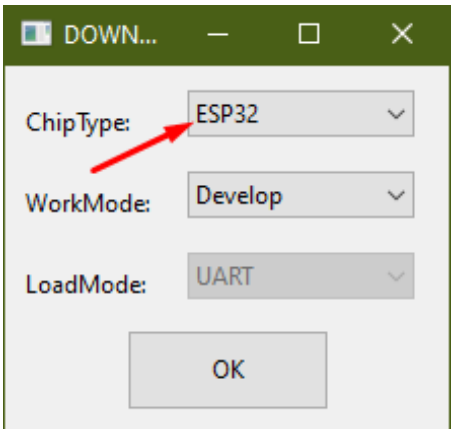
Install the following libraries in arduino:

Go to Tools → Manage libraries → (Search the library).

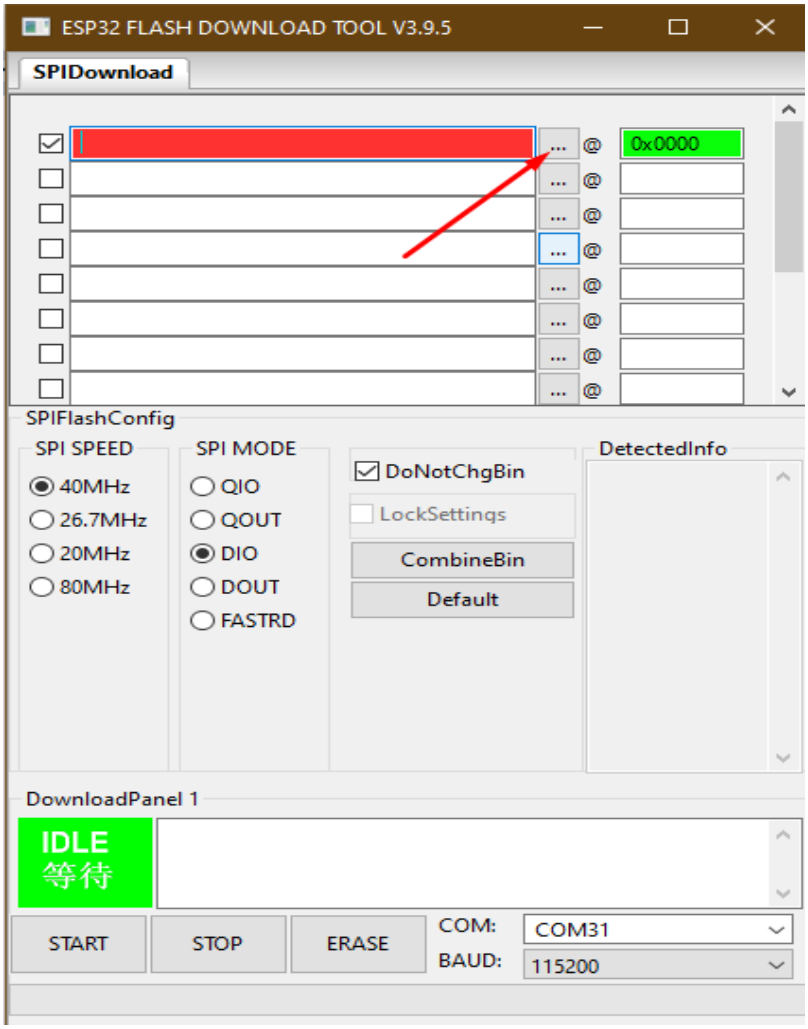
- ArduinoJSON(v7.2.1)
- HttpClient(v2.2.0)

2- Build Binaries:

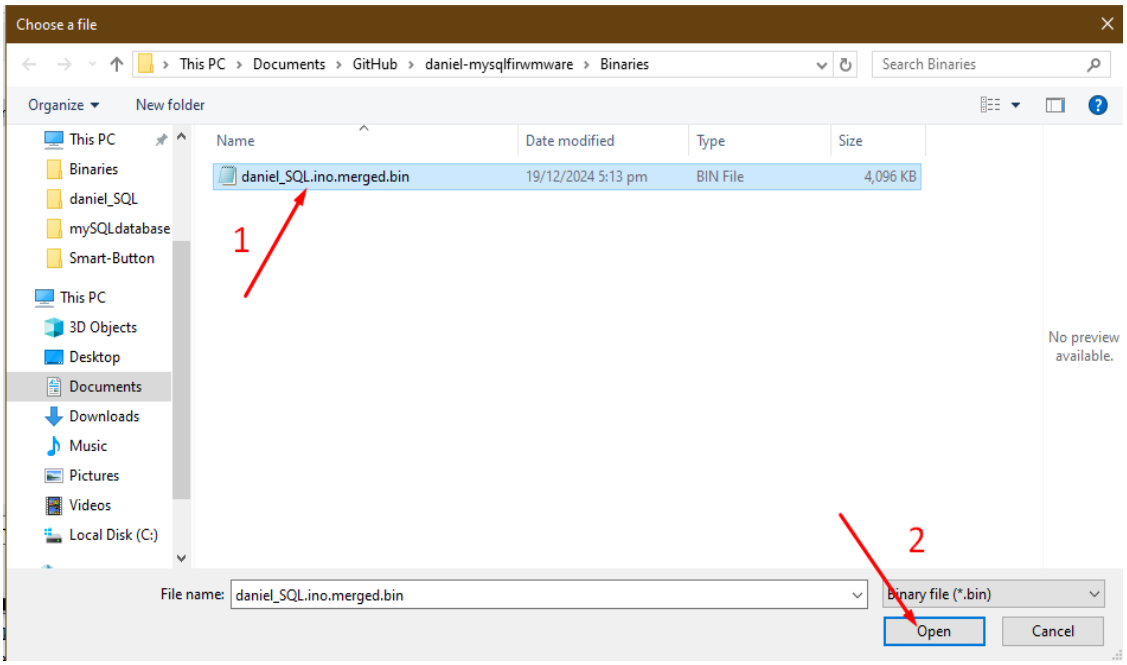
- Download the flash_download_tool from the bellow link and install.
  https://github.com/Makerfabs/esp_tool
- **After installation open the** flash_download_tool and setup the board.



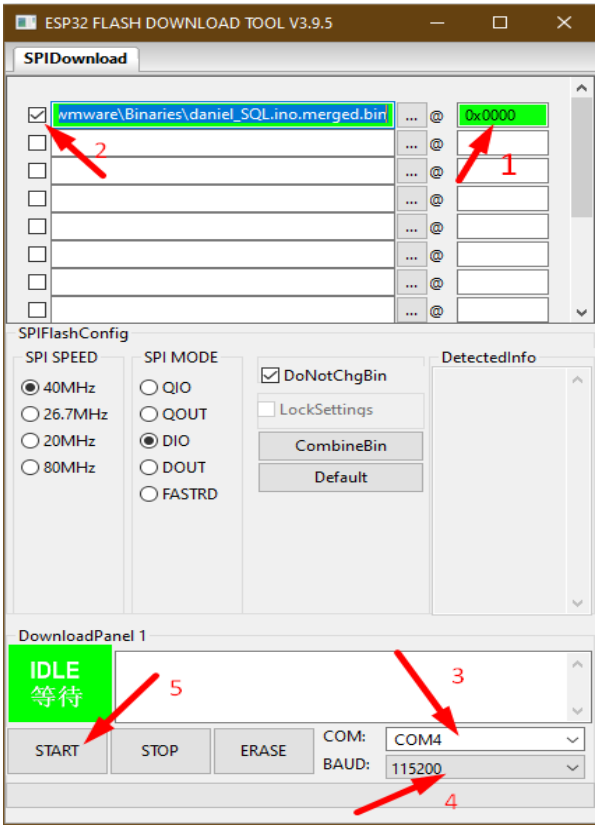
- Click on the three dots to open file browser.



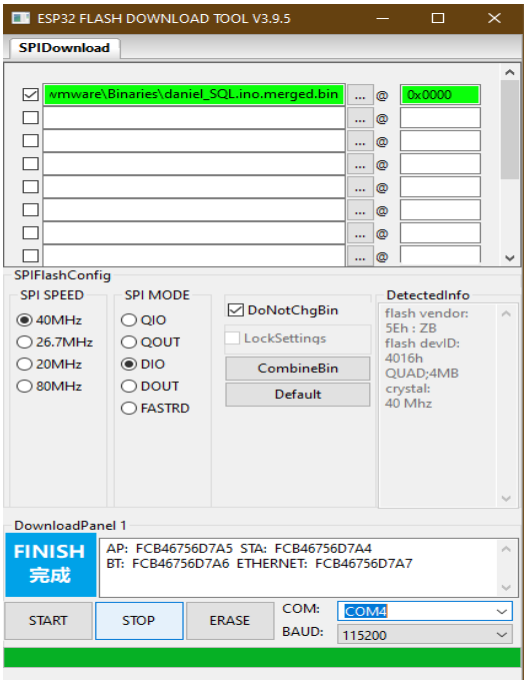
- Select the binary file from the directory and open.



- Make sure the address should **0x00.**
- Select the port and Baudrate the click on start.



- Binary file has been upload on ESP32.



- **NOTE:** Make sure same COM port shouldn't be open (like Serial Monitor in Arduino IDE) other wise COM FAIL error occured.

3- Setup DataBase and Server:

Step 1: Create a Database on MySQL

Ensure that MySQL Workbench is installed and configured. Update the `user` and `password` fields in `mysql_database.py` to match your MySQL Workbench credentials.

Configuration:

```
host="localhost",
user="root",              # Replace with your MySQL username
password="mysql123456",   # Replace with your MySQL password
database="esp32_data"
```

Database Setup:

1. Install the MySQL connector for Python:

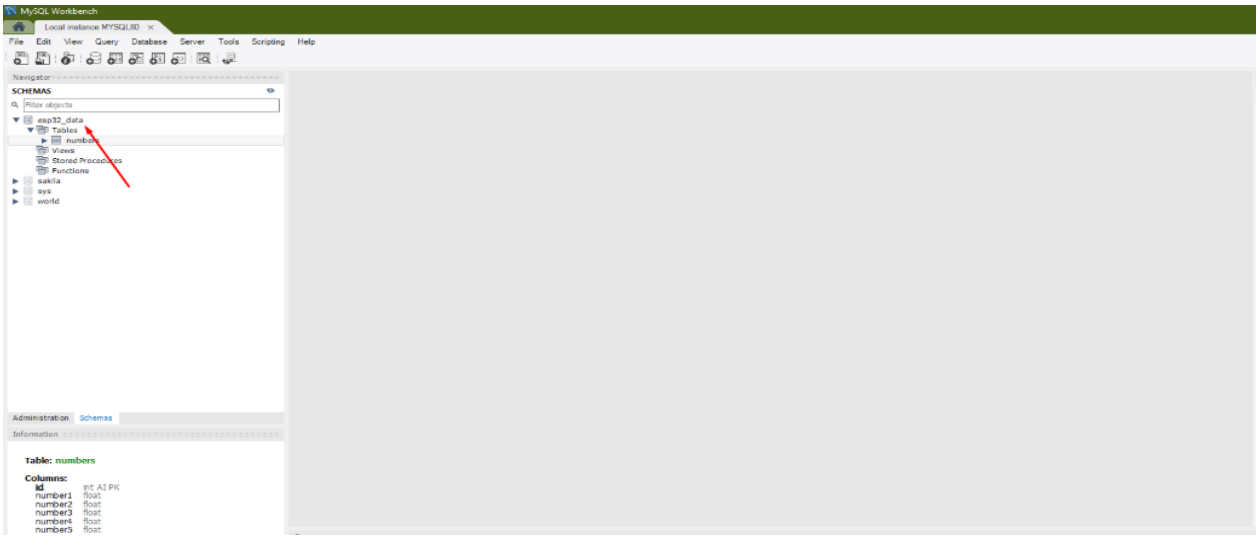
```
pip install mysql-connector-python
```

2. Run the database creation script:

```
python mysql_database.py
```

Verification:

- - Open MySQL Workbench and confirm the creation of the database `esp32_data`.



Step 2: Create a Server

Update the `user` and `password` fields in the server script to match your MySQL Workbench credentials.

Configuration:

```
host="localhost",
user="root",              # Replace with your MySQL username
password="mysql123456",   # Replace with your MySQL password
database="esp32_data"
```

Server Setup:

1. Install Flask:

```
pip install flask
```

2. Start the server:

```
python api.py
```

Verification:

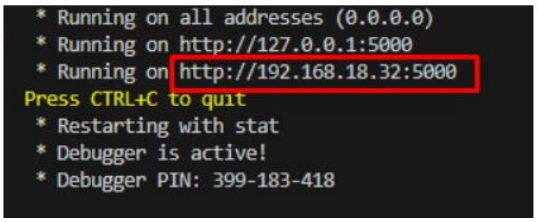
- The server should start and display a running message in the terminal.

```
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://192.168.18.32:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 399-183-418
```

Step 3: Update Firmware:

Ensure that both the server and ESP32 device are connected to the same network.





1. Compile the firmware using your preferred IDE.
2. Upload the compiled firmware to the ESP32 device.

Note:

- - Confirm network connectivity between the ESP32 and the server.

Resources:

Demo Video:

[Watch the Demo Video](https://www.loom.com/share/41fb0520eaf24f73900536a465b07a0f?sid=ff07c38c-4034-4e3b-81e7-210350594474)